# Across CES

## Share Functions as Distributions

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# Outline

1. The observation: CES shares are log-logistic
2. Economic content of the log-logistic parameters
3. Beyond CES: Theorem 1
4. Application: Burr XII

# What is a demand system?

Any demand system $\mathbf{x}(\mathbf{p}, Y)$ can be written as

$$x_j(\mathbf{p}, Y) = \frac{Y}{p_j} s_j(\mathbf{p}, Y),$$

where the **share functions** $s_j(\cdot)$ satisfy

$$s_j(\mathbf{p}, Y) \geq 0, \qquad \sum_{j=1}^{n} s_j(\mathbf{p}, Y) = 1.$$

This is an identity — it imposes no behavioral restrictions.

**Key point:** the share functions **are** the demand system. Everything that follows is about their shape.

# CES: the standard example

A representative consumer with CES utility over $n$ goods:

$$U = \left( \sum_{j=1}^{n} \alpha_j^{\frac{1}{\sigma}} x_j^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

Utility maximization yields demand $x_j = \frac{\alpha_j p_j^{-\sigma}}{\sum_k \alpha_k p_k^{1-\sigma}} Y$, so the share function is:

$$s_j(\mathbf{p}, Y) = \frac{\alpha_j p_j^{1-\sigma}}{\sum_{k=1}^{n} \alpha_k p_k^{1-\sigma}}$$

Positivity: immediate. Adding-up: immediate.

# CES shares as a function of own price

Fix good $j$ and hold $(\mathbf{p}_{-j}, Y)$ constant. Divide numerator and denominator by $\alpha_j p_j^{1-\sigma}$:

$$s_j(p_j;\, \mathbf{p}_{-j}, Y) = \frac{1}{1 + \underbrace{p_j^{\sigma-1} \sum_{k \neq j} \left(\frac{\alpha_k}{\alpha_j}\right) p_k^{1-\sigma}}_{\equiv\, B_{-j}\, p_j^{\sigma-1}}}$$

where $B_{-j} > 0$ collects everything that does not depend on $p_j$.

As a function of $p_j$ alone, this maps $\mathbb{R}_{++} \to (0, 1)$.

# The observation: CES shares are log-logistic

**CES share** (as a function of own price):

$$s_j(p_j) = \frac{1}{1 + B_{-j}\, p_j^{\sigma-1}}$$

**Log-logistic CCDF** (complementary CDF):

$$\bar{F}(x) = \frac{1}{1 + \left(\frac{x}{\beta}\right)^{\gamma}}$$

These are the same object. Set $x = p_j$, $\gamma = \sigma - 1$, $\beta = B_{-j}^{-1/(\sigma-1)}$.

No stochastic choice or random utility is introduced — the CDF structure is a **characterization** device.

# What do the log-logistic parameters mean?

The CES-to-log-logistic mapping: $s_j(p_j) = \bar{F}(p_j;\, \beta, \gamma)$ with

- $\gamma = \sigma - 1$ **(shape):** the elasticity of substitution minus one.
  - Governs how fast expenditure share drops as own price rises.
  - Higher $\sigma \Rightarrow$ steeper drop $\Rightarrow$ more substitutable goods.
  - Controls the **entire** shape: steepness, tails, hazard rate.
- $\beta = B_{-j}^{-1/(\sigma-1)}$ **(scale):** the competitive position of good $j$.
  - Pinned down by $(\mathbf{p}_{-j}, \boldsymbol{\alpha})$ — the prices and taste weights of all other goods.
  - The price at which good $j$'s share is exactly half.
  - Better competitors (lower $\mathbf{p}_{-j}$, higher $\boldsymbol{\alpha}_{-j}$) $\Rightarrow$ share drops sooner.

# The limitation of the log-logistic

In CES, $\gamma = \sigma - 1$ controls **everything** about the shape:

- The steepness of the share function near the median price.
- The thickness of the tails (how persistent shares are at extreme prices).
- The hazard rate profile: $\lambda(p_j) = \frac{\gamma}{\beta}\left(\frac{p_j}{\beta}\right)^{\gamma-1} \cdot s_j(p_j)$
  - $\gamma > 1$ ($\sigma > 2$): inverted-U hazard — demand sensitivity peaks then fades.
  - $\gamma \leq 1$ ($\sigma \leq 2$): monotone decreasing hazard.
  - No other options.

**The problem:** you cannot separate "how sensitive is reallocation near the median" from "how thick are the tails." One parameter does too much work.

# Generalizing CES

A family of demand systems **generalizes CES** if CES is recovered for some parameter values $\tilde{\boldsymbol{\kappa}}$.

**Theorem 1**

Suppose $\tilde{\mathbf{x}}(\mathbf{p}, Y, \boldsymbol{\kappa})$ generalizes CES. Then:

(i) Each generalized share $\tilde{s}_j(p_j; \boldsymbol{\kappa})$ is a generalization of the log-logistic (C)CDF.

(ii) If $\tilde{s}_j(\cdot)$ is **monotone** in $p_j$ and admits **limits** at zero and infinity, then $\tilde{s}_j(\cdot)$ is itself a (C)CDF.

The result is purely representational: no utility maximization or integrability required.

# What does the theorem ask for?

Part (ii) requires two conditions on the generalized share $\tilde{s}_j(p_j)$:

1. **Monotonicity:** $\tilde{s}_j$ is strictly decreasing in own price (CCDF) or strictly increasing (CDF).
   - In CES: always satisfied. Governed by $\sigma \gtrless 1$.
   - Economically: expenditure share moves in one direction as own price rises.

2. **Limits:** $\tilde{s}_j(p_j) \to 1$ as $p_j \to 0$ and $\tilde{s}_j(p_j) \to 0$ as $p_j \to \infty$ (for the CCDF case).
   - In CES: immediate from the $\frac{1}{1+(\cdot)}$ form.
   - Economically: as the price of a good approaches zero it gets all expenditure; an infinitely expensive good captures none.

These are mild conditions — they rule out pathological share functions, not interesting ones.

# The constructive direction

Theorem 1 has a converse:

$$\textbf{Choose a family of distribution functions} \implies \textbf{get a demand system}$$

that nests CES by construction.

The entire toolkit of distribution theory — monotonicity, tail behavior, hazard rates, parametric families — becomes available for demand system design.

Existing generalizations fit naturally: HSA and HDIA (Matsuyama & Ushchev, 2017/2021) can be rewritten as generalized log-logistic share systems.

# Burr XII: decoupling shape from tails

Replace the log-logistic CCDF with a **Burr Type XII** CCDF:

$$\tilde{s}_j(p_j) = \frac{1}{\left(1 + B_{-j}\, p_j^{\sigma-1}\right)^{\boldsymbol{k}}}$$

- At $\boldsymbol{k = 1}$: this is exactly CES (log-logistic).
- $\gamma = \sigma - 1$ still governs the central elasticity of substitution.
- $\boldsymbol{k}$ is a new parameter that controls **tail thickness**, independently of $\gamma$.

$k > 1$: Thinner tails — shares drop off **faster** at extreme prices.

$k < 1$: Fatter tails — shares are **stickier** at extreme prices.

$k = 1$: CES.

# What does $k$ buy you economically?

The Burr XII hazard rate is:

$$\lambda(p_j) = \frac{k\gamma}{\beta}\left(\frac{p_j}{\beta}\right)^{\gamma-1} \cdot \frac{1}{1+\left(\frac{p_j}{\beta}\right)^{\gamma}}$$

- $k$ scales the hazard rate without changing where it peaks.
- **Economically:** $k$ controls how persistent expenditure shares are when prices move far from their competitive level.
  - $k > 1$: demand reallocation **accelerates** in the tails. A firm that becomes very expensive loses share faster than CES predicts.
  - $k < 1$: demand reallocation **decelerates**. Brand loyalty, switching costs, or captive consumers keep share high even at extreme prices.
- CES forces $k = 1$: tail behavior is locked to the central elasticity.
  Burr XII lets you fit the center and the tails separately.

# Summary

1. CES share functions are **log-logistic distribution functions** in own price.
2. The log-logistic parameters have direct economic content:
   - $\gamma = \sigma - 1$: elasticity of substitution (shape).
   - $\beta$: competitive position relative to all other goods (scale).
3. Any demand system nesting CES inherits this structure. If shares are **monotone** with correct **limits**, they are distribution functions (Theorem 1).
4. The identification is **constructive**: choose a distribution family, get a demand system.
5. **Burr XII** demonstrates the payoff: one extra parameter ($k$) decouples tail behavior from the central elasticity, something CES cannot do.